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1 Appendix

1.1 Implementation details of the baseline models

Setup. We implemented all the models using the encoder-decoder architecture. The encoders are different baseline models (listed below). The encoder takes as input the given story (paragraph) $p = (p_1, p_2, \dots)$ and produces the representation of the story. In all the models, the decoder is implemented as a 2-layer MLP which takes as input the concatenated representation of the story and the embedding of the entities (for which the relationship is to be predicted) and returns a softmax distribution over the relation types. We now describe the different baseline models (encoders) in detail:

LSTM (Hochreiter and Schmidhuber, 1997): The input paragraph is processed by a two-layer Bidirectional LSTM and the hidden state corresponding to the last time-step is used as the representation of the story.

LSTM+attention (Cho et al., 2014): Similar to LSTM, but instead of using just the hidden state at the last timestep, the model computes the attention-weighted mean of the hidden state at all time steps to use as the representation of the story.

Relation Networks - RN (Santoro et al., 2017): A relation module (implemented as an MLP) is used alongside the LSTM to learn pairwise relations among all the pairs of sentences. These relation representations are the output of the relational module. Our input data is prepared as a batch of $\text{sentences} \times \text{words}$. Each sentence is fed to the LSTM, followed by a pooling (e.g. mean, max) over all hidden states of each sentence to generate the sentence embeddings. The query embeddings are no longer needed in the decoder since they have been incorporated by the relational module when learning relations between sentences.

MAC (Compositional Attention Network) (Hudson and Manning, 2018): A MAC cell is similar to RN, but it also contains a *control* state c and *memory* state m and can iterate over the input several times. The number of iterations is a hyperparameter. Just like RN, MAC is added behind the LSTM. In each iteration, the model attends to the embeddings of the query entities to generate the current control c_i . Another attention head over c_i and all hidden outputs of LSTM is used to distill the new information r_i . In the end, a linear layer is used to generate the new memory m_i by combining r_i and

m_{i-1} . The final memory state gives the representation of the story. This model is the state-of-the-art model for the CLEVER task.

BERT (Devlin et al., 2018): We adapt BERT pre-trained language model to our task. Specifically, we use two variants of BERT - the vanilla 12-layered frozen BERT with pre-trained embeddings, and BERT-LSTM, where a one-layer LSTM encoder is added on top of pretrained BERT embeddings. BERT encodes the sentences into 768-dimensional vectors. To ensure that BERT does not treat the entities as unknown tokens (and hence producing the same representation for all of them), we represent the entities with numbers in the vanilla BERT setup. In BERT-LSTM, we replace the entity embeddings by our entity embedding lookup policy (Refer Appendix 1.5). In both the cases, we use a simple two-layer MLP decoder which takes as inputs the pooled document representation and query representations and produces the softmax distribution over the relations.

Graph Attention Network (GAT) (Veličković et al., 2018): Entity (modelled as nodes in the graph) representations are learned by using the GAT Graph Neural Network with attention-based aggregation over the neighbor nodes. We modify the GAT architecture by attending over each node v_j in the neighborhood of v_i by concatenating the edge representation $e_{i,j}$ to the representation of v_i .

Relational Recurrent Network (RMC) (Santoro et al., 2018): We also implemented RMC, a recently proposed model for relational reasoning. It works like an RNN, processing words step-by-step, except that a memory matrix ($\text{num_slots} \times \text{mem_size}$) is added as the hidden state. The relational bias is extracted in each step by using self-attention over the concatenation of memory matrix and the word input in the current step. The final memory matrix is the representation of the story. Our implementation is based on another open source implementation.⁶ We noticed that the performance of the model is significantly less across all the tasks by a large margin. Till the time of the submission, we could not verify whether this subpar performance is due to buggy implementation of the code or due to some unexplored hyperparameter combination. Hence we decided not to include the results corresponding to this model in the empirical evaluation. We will continue working on verifying the implementation of the model.

⁶<https://github.com/LOSG/relational-rnn-pytorch>